

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Technology (CSE)

QUIZ #1 (SET-B)

DURATION: 25 MINUTES

SUMMER SEMESTER, 2024-2025

FULL MARKS: 15

CSE 4621: Machine Learning

Programmable calculators are not allowed.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

1. For a simple dataset $\{(1, 1.75), (2, 3.22), (3, 1.67)\}$, calculate one step of gradient update, if $h_{\theta}(x^{(i)}) = \theta_0 + \theta_1 x^{(i)}$ and initially, $\theta_0 = 0, \theta_1 = 0$ and $\alpha = 1$. 10
(CO1)
(PO1)

Solution:

Gradient for θ_0 : $\frac{\partial J(\theta)}{\partial \theta_0} = \frac{1}{3}(-1.75 - 3.22 - 1.67) = -2.213$

Gradient for θ_1 : $\frac{\partial J(\theta)}{\partial \theta_1} = \frac{1}{3}(-1.75 - 6.44 - 5.01) = -4.4$

Thus, $\theta_0 = 0 - \alpha(-2.213) = 2.213$ and $\theta_1 = 0 - \alpha(-4.4) = 4.4$

Rubric:

- 5 points for correctly finding the sub-terms.
- 5 points for correctly finding the solution.

2. If a loss function is non-convex and has multiple shallow local minimum values, should we use batch or stochastic gradient descent? 5
(CO1)
(PO1)

Solution:

SGD's noisy gradient estimates introduce random perturbations or noise at each step. If a local minimum is shallow (its basin is narrow and the loss advantage over nearby regions is small), the noise may be large enough to kick the solution out of it, allowing exploration of potentially better regions.

Rubric:

- 5 points for correctly explanation.